EM Portfolio Investment Reversal Prediction

A Logit and Random Forest Early-Warning Comparison

Name: Mohamed Ammar

1. Introduction

Emerging market economies are periodically exposed to sharp reversals in non-resident portfolio investment, in which capital that had been flowing in withdraws abruptly, often with real consequences for exchange rates, credit conditions, and output. Anticipating these episodes, even imperfectly, is of direct interest to policymakers and investors alike, and has motivated a long-standing literature on early-warning models for capital flow reversals and related crisis events.

This paper builds and evaluates a quarterly, cross-country model of portfolio investment reversals across 23 emerging market economies from 2000Q1 onward. The research question is straightforward: using a standard set of global ("push") and country-specific ("pull") factors, observed with a one-quarter lag relative to the outcome, can a reversal be predicted ahead of time using only information available before it occurs? Two modeling approaches are compared — logistic regression, the conventional choice in this literature, and a Random Forest, a more flexible machine-learning alternative that has seen more limited use in this specific setting.

The emphasis throughout is on genuine out-of-sample evaluation. Reversals are defined using an expanding-window 10th-percentile threshold that only uses each country's own history up to that point, and both models are evaluated using a forward-chaining scheme that trains only on data prior to the period being predicted. This is a deliberate response to a risk common in this kind of modeling: that a model can appear highly predictive in-sample while adding little genuine forecasting value once evaluated honestly on data it has not seen.

The remainder of this paper proceeds as follows. Section 2 describes the data and variable construction. Section 3 sets out the two models and the specification choices made throughout. Section 4 reports results, including a single train/test split and the forward-chaining evaluation, along with variable importance. Section 5 reflects critically on the limitations of this approach and what would be done differently. Section 6 concludes.

2. Data

2.1 Sample and research design

The analysis draws on a quarterly panel of 23 emerging market (EM) economies covering 2000Q1 onward: Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Hungary, India, Indonesia, Korea, Malaysia, Mexico, Nigeria, Peru, Philippines, Poland, Russia, Saudi Arabia, South Africa, Thailand, Türkiye, and Ukraine. Three additional EM economies commonly included in the capital-flows literature — Lebanon, the UAE, and Venezuela — were excluded due to unavailable quarterly GDP data, which enters the model both as a predictor and as a scaling denominator.

A single-country early warning design was initially considered but set aside on sample-size grounds: even 25 years of quarterly history for one country yields only ~100 observations, insufficient for a rare-event outcome. A cross-country panel was adopted instead to multiply the effective sample size while retaining quarterly granularity.

The dependent variable is sourced from the IIF EM Portfolio Flows Tracker rather

than official IMF BPM6 balance-of-payments data, because BPM6 coverage has severe start-date gaps for several sample countries (e.g., Egypt from 2011Q1, Nigeria from 2014Q1). IIF data are shown by Koepke and Paetzold (2020) to be broadly consistent with official BPM6 flows; this was independently verified for Egypt against Central Bank of Egypt data.

2.2 Variable definitions, sources, and rationale

Variable selection reflects both theoretical relevance and practical data availability across the full panel; some variables that might otherwise be preferable (e.g., higher-frequency sentiment or capital-control indices) could not be sourced consistently across all 23 countries and were excluded on that basis. **Table 1** summarizes all variables, their construction, source, and rationale for inclusion.

Dependent variable

Variable	Description	Data source	Rationale
Reversal	Binary: 1 if quarterly non-resident portfolio investment falls at or below the expanding-window 10th percentile of that country's own history to date	IIF EM Portfolio Flows Tracker (Farnham & Tiftik, 2018)	No-look-ahead-bias construction; expanding-window design follows Calvo et al. (2004)

Global ("push") factors

Variable	Description	Source	Rationale
`VIX`	CBOE Volatility Index, quarterly average	FRED	Global risk aversion; central to the "global financial cycle" (Rey, 2013)
`Fed_Funds_Rate`	US federal funds rate, quarterly average	FRED	Center-country monetary policy (Rey, 2013)
`Brent_Oil_Price`	Brent crude, quarterly average	FRED	Commodity/terms-of-trade shock
`us_dollar_index`	US dollar effective exchange rate	IMF	Broad dollar strength / global liquidity channel

Country-specific ("pull") factors

Variable	Description	Source	Rationale
exchange_rate	LCU per USD	IMF	Currency pressure (Calvo et al., 2004)
reserves_pct_gdp	FX reserves ÷ trailing 4-quarter nominal GDP, ×100	IMF	Reserve adequacy as a share of economic size (Frankel & Saravelos, 2012); same stock-flow annualization as market_cap_gdp_ratio
gdp_growth_yoy	Real GDP growth, YoY	IMF (Egypt: national data)	Isolates real momentum from inflation; YoY nets out seasonality in an unadjusted series
m2_growth_yoy	M2 growth, YoY	IMF	Domestic liquidity conditions
market_cap_gdp_ratio	Quarter-end market cap / trailing 4-quarter GDP, %	WFE; GDP from IMF	Corrects stock-flow mismatch
Interest_Rate	Policy rate	IMF, BIS-supplemented for 4 countries	Monetary stance / US rate differential
Stock_Return_Percent	Quarter-end index % change	National exchanges	Domestic sentiment/performance
Stock_Volatility	SD of 3 within-quarter monthly % changes	National exchanges	Domestic risk, distinct from VIX
current_account_pct_gdp	CA balance / nominal GDP	IMF (constructed)	External financing need

All country-specific predictors are lagged one quarter relative to the dependent variable to avoid contemporaneous reverse-causality.

reserves_pct_gdp is constructed as FX reserves divided by trailing 4-quarter nominal GDP, using the same stock-flow annualization logic applied to market_cap_gdp_ratio. Two data errors were identified and corrected during construction: a units mismatch (reserves reported in millions against GDP in raw currency units) and an initial log-level specification that did not match the intended ratio. The corrected series was verified against documented reserve figures for a subset of countries — for example, China's 1999Q4 value of 14.0% matches the historically documented range of roughly 13-14% — and all countries now fall within a plausible 5-22% range, versus an implausible 22-77% range before correction.

Contrary to the expected sign, reserves_pct_gdp shows a consistent *positive* coefficient across specifications (higher reserves associated with higher reversal probabil-

ity). This is interpreted as reverse causality — defensive reserve accumulation occurring in response to elevated reversal risk, rather than reserves preventing reversals — and is discussed further in Section 4 (Results) and Section 5 (Critical Reflection), rather than treated as a data error, since the corrected and independently verified series still exhibits this pattern.

2.3 Dependent variable: reversal indicator

A reversal is defined as a quarter in which portfolio investment falls at or below the 10th percentile of that country's own history *up to and including that quarter* — an expanding-window threshold recalculated at every point in time. This avoids look-ahead bias inherent in a static full-sample threshold, which would implicitly use knowledge of future crises (e.g., 2008, 2020) to calibrate what counted as "extreme" in earlier periods. A 12-quarter warm-up is required before flagging begins, balancing statistical stability against the panel's data constraints for shorter-history countries.

This percentile-based threshold departs from the more commonly used standard-deviation-based definition in the sudden-stop literature (e.g., a fall of 1 to 2 standard deviations below the country's mean; Calvo et al., 2004). A standard-deviation threshold assumes portfolio flows are reasonably close to normally distributed, an assumption that is questionable for a variable prone to fat tails and sharp, asymmetric drops during crisis episodes; a percentile threshold instead makes no distributional assumption, and is inherently more robust to the extreme observations it is trying to identify, since it does not depend on a mean and standard deviation that the extreme events themselves would distort. A fixed percentile also fixes the reversal event rate directly (approximately 10% of each country's own history, by construction), which is useful for a supervised classification design and keeps the class balance broadly comparable across countries with very different levels of flow volatility — something a fixed standard-deviation cutoff would not guarantee, since more volatile countries would mechanically generate more or fewer flagged events depending on how volatility itself evolves over time.

2.4 Data cleaning and coverage

Coverage is materially incomplete for several countries even after construction — 64.4% of country-quarter observations (1,645 of 2,553) have complete data across all 13 predictors, with the weakest coverage concentrated in Saudi Arabia (25.2%), Russia (35.1%), Malaysia (36.9%), and Nigeria (37.8%), and Ukraine excluded entirely from specifications requiring the full predictor set (0.0% complete). This is disclosed as a limitation (Section 5) rather than addressed through imputation.

3. Methodology

3.1 Logistic Regression (Logit)

The first method applied is logistic regression (logit), the default modeling choice in the early-warning-system (EWS) literature since it directly estimates the probability

of a binary outcome — here, whether a reversal occurs in a given quarter — as a function of the 13 lagged predictors. The logit model has the advantage of directly interpretable coefficients: the sign indicates the direction of a variable's association with reversal risk, and the magnitude (after standardization) allows rough comparison of relative importance across predictors. Some EWS studies extend this binary framework to a multinomial logit with four states — typically tranquil, pre-crisis, crisis, and post-crisis periods — allowing the model to distinguish the build-up phase from the crisis itself rather than pooling all non-reversal quarters into a single category (Bussière & Fratzscher, 2006). This study retains the binary specification because the object of interest is the reversal event itself — whether a sharp portfolio outflow occurs in a given quarter — rather than distinguishing among finer-grained states of the capital-flow cycle; the multinomial extension is noted as a possible direction for future work (Section 5). Predictors are standardized prior to estimation, and out-of-sample performance is evaluated using an expanding-window forward-chaining scheme, refitting the model at each step on all data strictly prior to the evaluation period, consistent with the no-look-ahead-bias design used throughout this study (Section 2.3). Given the low reversal event rate, model performance for the logit specification is assessed primarily using the F1-score and AUROC, alongside average out-of-sample prediction accuracy, rather than raw accuracy alone, which would be inflated by the majority "no reversal" class.

3.2 Random Forest

The second method applied is CART (Classification and Regression Trees) for binary classification, ensembled into a random forest, following Breiman (2001). CART works by recursively splitting the sample: at each node, observations are divided into two groups based on the variable and threshold that best separate reversals from non-reversals. This process continues until a terminal node is reached, where all observations are assigned the majority class. A single CART tree tends to overfit and produces unstable predictions out-of-sample. A random forest addresses this by growing many trees and averaging their predictions, which reduces variance. Two sources of randomness make the trees different from one another: each tree is trained on a resampled version of the training data, and at each split, only a random subset of predictors is considered rather than the full set. This decorrelation across trees is what allows the ensemble to outperform any single tree in out-of-sample prediction. Unlike the primary logit specification (Section 3.1), the Random Forest specification uses class-weight balancing throughout, reweighting the loss function to counteract the low reversal event rate; this asymmetry between the two models' handling of class imbalance, and its consequences, are discussed in Section 4.2 and revisited in Section 5.2. As with the logit model, performance is assessed using the F1-score and AUROC alongside average out-of-sample prediction accuracy, given the class imbalance in the reversal outcome.

Random forests remain relatively rare in the early-warning-system literature, where the dominant approaches are the signaling method and logistic regression, both of which are more directly interpretable (Jarmulska, 2022) — a logit coefficient has a clear sign and magnitude, and a signaling indicator crosses a defined threshold. A random forest, by contrast, is often treated as a "black box": prediction accuracy can improve, but understanding *why* the model predicts a reversal becomes harder, since

the outcome depends on hundreds of trees rather than one transparent equation. To partially address this, Breiman's variable importance measure is used, which quantifies how much each predictor contributed to reducing classification error across all trees in the forest. This does not fully restore the interpretability of a logit coefficient, but it offers a way to identify which of the 13 predictors the model relies on most, and to compare this ranking against the signs and significance obtained from the logit specification.

3.3 Specification choices common to both models

Both models are estimated without country fixed effects, relying instead on the time-varying push and pull predictors to carry the model's signal. Country fixed effects would absorb cross-country variation into country-specific intercepts, which would change what the model's predictions depend on; this trade-off, and whether including fixed effects would be preferable, is left as a direction for future work (Section 5.4).

Similarly, neither model includes dummy variables for specific quarters, including quarters corresponding to widely recognized global shocks (e.g., 2008Q4, 2020Q1-Q2). Such dummies would only be informative if the model already "knew," at estimation time, that the quarter in question was extraordinary — information that, under the forward-chaining evaluation design used here, is precisely what the model is being asked to predict rather than assume. Including a dummy for an extraordinary quarter that falls within the test period would mean the test observation had already been labeled as unusual before the model made its prediction, which is a direct form of future information leakage rather than a genuine out-of-sample forecast. For this reason, extraordinary periods are left to be identified by the model indirectly, through the behavior of the global push factors (VIX, Fed_Funds_Rate, Brent_Oil_Price, us_dollar_index) that are elevated or extreme during such episodes, rather than through an explicit time dummy.

4. Results

4.1 Baseline for comparison

Before turning to model performance, it is useful to establish what a naive, uninformed prediction would achieve. Across the full sample, the reversal event rate is approximately 16.6%, so a model that always predicts "no reversal" would be correct roughly 83-84% of the time on the sample as a whole (naive baseline accuracy = 1 - event rate). This full-sample figure is not identical to the naive baseline for any specific test window, however: the event rate — and therefore the naive baseline — can differ across sub-periods. For example, the 2021Q2-2025Q4 test window used in Section 4.2 has a higher event rate (19.3%), giving a naive baseline of 80.7% for that specific split (Table 2), while the 2010Q1-2025Q4 walk-forward evaluation in Section 4.3 has a naive baseline of 83.5%, closer to the full-sample figure. Each result below should be read against the naive baseline reported for that specific evaluation, not against the full-sample figure.

4.2 Single time-split results

Both models were first evaluated on an identical time-based 80/20 split: all observations before 2021Q2 were used for training, and all observations from 2021Q2 onward were held out for testing. This ensures the test period is strictly later in time than the training period for both models, so the two sets of results below are directly comparable. The Random Forest specification uses class-weight balancing (`class_weight="balanced"`); the logit specification reported below does not, for reasons discussed at the end of this section.

Table 2. Single time-split test performance (train: 2002Q4-2021Q1; test: 2021Q2-2025Q4)

Metric	Logit	Random Forest (balanced)
Accuracy	0.723	0.701
F1-score	0.119	0.143
AUROC	0.407	0.370
Naive baseline accuracy	0.807	0.807

Both models underperform relative to what a purely uninformed prediction would achieve on this specific test window: accuracy for both falls below the 80.7% naive baseline, and AUROC for both falls below 0.5, the threshold at which a model provides no better-than-random ranking of reversal risk. This is a striking and important result: on this particular four-year holdout period, neither model demonstrates genuine discriminative power. Random Forest achieves a marginally higher F1-score than logit (0.143 vs. 0.119), but a lower AUROC (0.370 vs. 0.407), so no model can be declared unambiguously superior on this split.

This weak result should not be over-interpreted in isolation, however, since a single test window covering only four years captures a specific, and possibly unrepresentative, stretch of history. Section 4.3 addresses this by re-evaluating both models using a forward-chaining (expanding-window) cross-validation scheme across many quarterly test windows, rather than a single split.

A note on class-weight balancing. A balanced logit was also tested and raised recall sharply (0.055 to 0.807) and F1 (0.119 to 0.294), but at a severe cost to accuracy, which fell to 0.252 — below the 0.807 naive baseline — since the model flagged 278 of 321 observations as reversals, only 62 of which were true. This precision profile (0.180) would generate far more false alarms than genuine warnings, and adopting it requires a policy judgment about relative costs (Section 5.4) this paper does not make. AUROC was essentially unchanged (0.410 vs. 0.407), confirming balancing shifts the classification threshold rather than improving the model's ranking quality. The unbalanced specification is therefore retained as the primary result.

4.3 Walk-forward (forward-chaining) results

To obtain a more robust measure of out-of-sample performance, both models were re-evaluated using an expanding-window forward-chaining scheme: starting from 2010Q1, each quarter is predicted using a model trained only on data from all prior quarters, and the model is refit before each new quarter's prediction. This produces

64 separate out-of-sample evaluations per model (2010Q1 through 2025Q4), rather than a single train/test split, and better reflects how a real-time forecaster would use the model.

Table 3. Walk-forward evaluation results (2010Q1-2025Q4, 64 quarters, 1,220 total predictions)

Metric	Logit	Random Forest (balanced)
Average accuracy (mean across quarters)	0.824	0.714
Pooled accuracy	0.818	0.715
Pooled AUROC	0.534	0.550
Pooled F1-score	0.090	0.179
Pooled recall	0.055	0.189
Naive baseline accuracy	0.835	0.835

Two patterns stand out. First, both models' walk-forward AUROC (0.534 and 0.550) is modestly above 0.5, indicating some genuine, if weak, discriminative signal — a meaningfully better result than either model's single-split AUROC in Table 2. This suggests the single time-split result in Section 4.2 reflected a particularly difficult four-year test window rather than a complete absence of predictive signal in the underlying approach.

Second, the two models trade off differently between accuracy and the ability to actually catch reversals. Logit's average accuracy (0.824) is higher than Random Forest's (0.714), but this is driven largely by logit defaulting to the majority "no reversal" prediction more often: logit's recall is only 0.055, meaning it correctly identifies just over 5% of true reversal events, compared to Random Forest's recall of 0.189 — over three times higher. Random Forest's F1-score (0.179) is roughly double logit's (0.090), indicating it does a meaningfully better job of balancing correct identification of reversals against false alarms, even though its overall accuracy is lower. As discussed in Section 4.2, applying class-weight balancing to the logit specification raises its walk-forward F1-score to 0.284 and recall to 0.592 — now exceeding Random Forest on both metrics — but at the cost of average accuracy falling to 0.499, below the naive baseline. This confirms, on the walk-forward evaluation, the same trade-off identified in Section 4.2, and the same reasoning for not adopting the balanced specification as the primary reported result applies here.

Quarter-by-quarter, both models struggle most during periods of acute global stress. The COVID-19 shock (2020Q1) produces the sharpest single-quarter accuracy drop for both models (logit: 0.429; Random Forest: 0.333), consistent with the difficulty of predicting a shock whose timing and severity had no close historical precedent in the training data available at that point.

Two further patterns are visible once accuracy is examined quarter by quarter alongside the number of true reversals in each quarter (n_reversals).

First, Random Forest performs markedly worse than logit in the earliest walk-forward quarters. Restricting to 2010-2011, Random Forest's mean accuracy is only 0.468, compared to 0.749 from 2012 onward — a substantially larger early-period penalty than logit shows over the same years. This is consistent with Random Forest requiring a larger training sample to fit stably: with only 2002Q4-2009Q4 available as training data at the start of the walk-forward evaluation, the expanding window

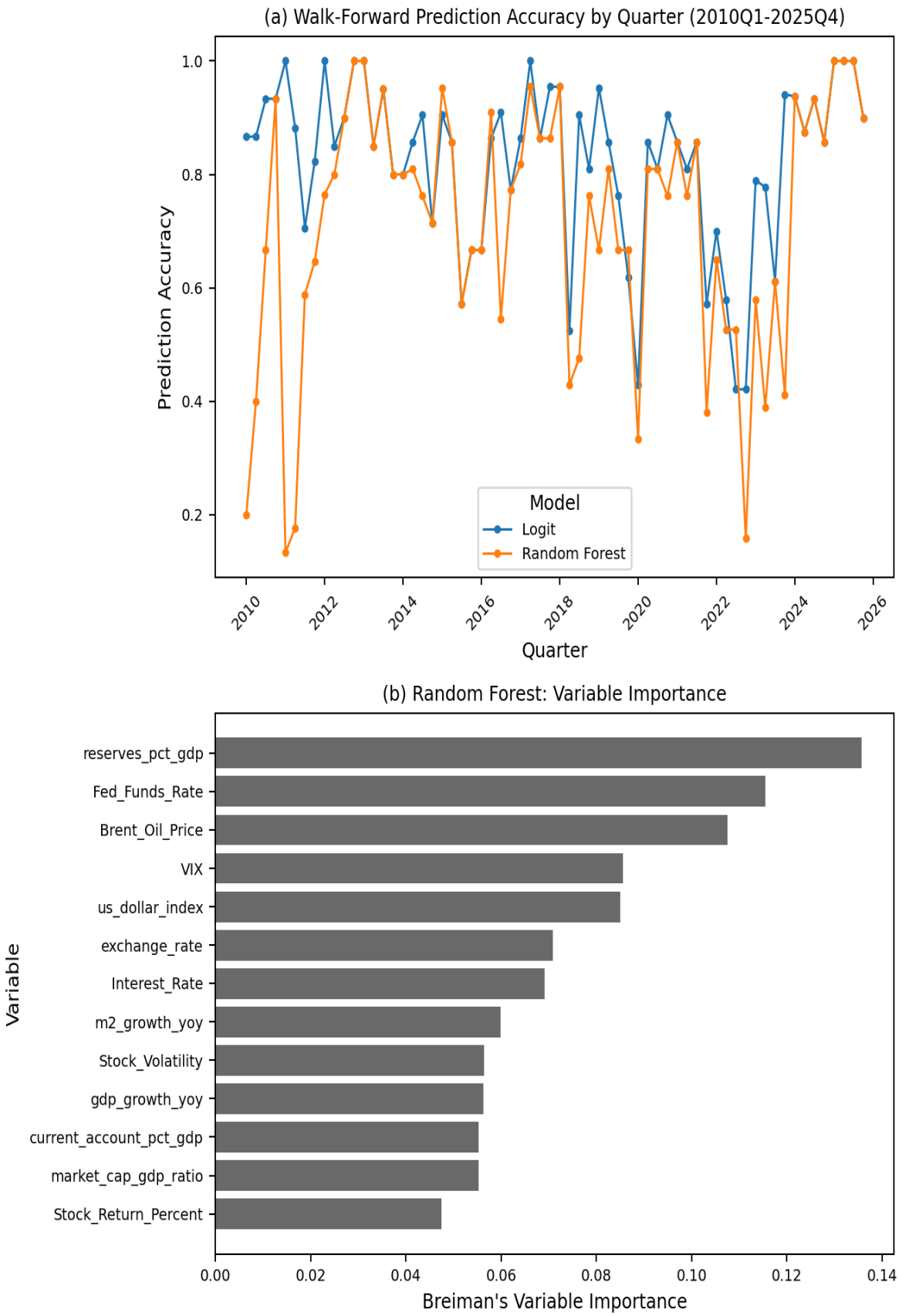


Figure 1: Walk-Forward Accuracy Comparison (top) and Random Forest Variable Importance (bottom)

has not yet accumulated enough observations for the forest's many decision splits to generalize well, whereas logit's much smaller number of estimated parameters (13 coefficients) can be fit reasonably even on a shorter history.

Second, accuracy in quarters with very few true reversals should be interpreted with caution. In quarters with zero or one reversal, mean accuracy is high for both models (0.934 for logit, 0.813 for Random Forest), but this partly reflects how easy it is to be "mostly correct" when there is at most one reversal to identify among 14-22 observations, not necessarily that the model successfully flagged that one case. For example, in 2019Q1 (one true reversal among 21 observations), logit achieves 0.952 accuracy and Random Forest 0.667 — yet in both cases, the single actual reversal may still have gone undetected, with the score driven almost entirely by correctly predicting "no reversal" for the remaining observations. By contrast, in quarters with five or more true reversals, mean accuracy falls to 0.615 (logit) and 0.584 (Random Forest) — a substantially lower figure that better reflects the models' genuine difficulty in identifying reversals when they are more frequent. This asymmetry is the same phenomenon underlying the F1/recall discussion above: headline accuracy in low-event quarters can mask a model's failure to catch the very events it is designed to warn about, and is itself a further argument for why F1 and recall, rather than accuracy alone, are the more informative metrics for this application (Sections 3.1-3.2).

4.4 Variable importance and coefficient comparison

Table 4. Random Forest variable importance (Breiman importance, n_estimators=200)

Rank	Variable	Importance
1	`reserves_pct_gdp`	0.136
2	`Fed_Funds_Rate`	0.116
3	`Brent_Oil_Price`	0.108
4	`VIX`	0.086
5	`us_dollar_index`	0.085

See panel (b) of the figure above for the Random Forest variable importance ranking.

The two models broadly agree that global push factors — particularly the US federal funds rate, oil prices, and the dollar index — carry substantial explanatory weight in the Random Forest ranking, consistent with the "global financial cycle" framing (Rey, 2013) motivating their inclusion. `reserves_pct_gdp` is the single most important predictor in the Random Forest model.

Separately, several logit coefficients run counter to the sign that standard early-warning theory would predict for that variable — most notably `reserves_pct_gdp`, which carries a positive coefficient (higher reserves associated with *higher* reversal probability, the opposite of the conventional buffering role reserves are expected to play), but also `VIX` and `Fed_Funds_Rate`. Rather than treating this here as a variable-by-variable interpretation exercise, these sign reversals are addressed as a modeling limitation in Section 5 (Critical Reflection), alongside the reverse-causality explanation already discussed for `reserves_pct_gdp` in Section 2.2.

4.5 Summary of findings

Taken together, the results support three main conclusions. First, both models show weak but genuine out-of-sample discriminative power under a properly time-ordered evaluation (walk-forward AUROC modestly above 0.5), but this signal is not strong or stable enough to be robust to the specific time window chosen for a single train/test split, where both models actually underperform the naive baseline. Second, Random Forest and logit make different trade-offs: logit achieves higher raw accuracy by predicting the majority class more often, while Random Forest achieves a better balance between catching reversals and avoiding false alarms, which is arguably more relevant for an early-warning application; class-weight balancing was found to improve logit's recall and F1-score substantially (Sections 4.2-4.3), but was not adopted as the primary specification given the severe accuracy cost this entailed, and is instead noted as a direction for further development (Section 5). Third, both models depend heavily on global push factors, and Random Forest's importance ranking corroborates the centrality of `reserves_pct_gdp` as a predictor, even though Random Forest's variable importance is unsigned and does not itself indicate direction. The logit coefficient for this variable carries a sign contrary to theoretical expectations, a puzzle discussed further in Section 5.

5. Critical Reflection

5.1 Sample and data limitations

The panel excludes three emerging-market economies commonly included in the capital-flows literature — Lebanon, the UAE, and Venezuela — solely because consistent quarterly GDP data was unavailable for them (Section 2.1). Lebanon and Venezuela in particular experienced significant crises within the sample period, so this exclusion is a real limitation: the model's already-weak out-of-sample performance may still overstate how well it would generalize to the countries where an early-warning system matters most.

Coverage is also materially incomplete: only 64.4% of country-quarter observations have complete data across all 13 predictors, concentrated in a handful of countries — Saudi Arabia (25.2%), Russia (35.1%), Malaysia (36.9%), and Nigeria (37.8%) have the weakest coverage, and Ukraine has 0% coverage and is excluded entirely (Section 2.4). No imputation was applied, to avoid introducing fabricated relationships into a rare-event outcome, but the effective sample is smaller and less representative than the headline country count suggests. Ukraine's blocking variable was never identified, an unresolved issue rather than a resolved one.

More time should have been allocated to data collection: higher-frequency data (monthly, from sources such as the IIF or EERF) was not accessible within this project's constraints and might have eased the coverage gaps and the warm-up requirement. It is also unclear whether the missing-data pattern reflects a genuine constraint or an avoidable gap in the sources used — interest rates, for example, are typically easy to source. Either way, the ability to fill gaps for some countries but not others risks a sample that is not missing data at random, but biased toward more data-accessible countries — separate from, and in addition to, the Lebanon/UAE/Venezuela exclusion above.

5.2 Modeling failures and unresolved puzzles

Counter-theoretical coefficient signs. Several logit coefficients carry a sign opposite to what standard early-warning theory would predict — most notably `reserves_pct_gdp` (Section 4.4), which is also the single most important predictor in the Random Forest model, but also `VIX` and `Fed_Funds_Rate`. Rather than a single-variable curiosity, this points to a broader issue with how these predictors relate to reversal risk that this paper does not resolve — for example, whether the relationships are genuinely non-monotonic, whether reverse causality is at play for some variables, or whether multicollinearity among the predictors is distorting individual coefficients. This is left as a specific, unresolved failure of interpretation for future work rather than an incidental footnote.

The balancing trade-off. Class-weight balancing was tested for the logit specification and found to improve recall (from 0.055 to 0.592 on the walk-forward evaluation) and F1-score (from 0.090 to 0.284), but at a cost to overall accuracy (falling to 0.499, below the naive baseline of 0.835; Section 4.2-4.3). This paper reports the unbalanced specification as its primary result, since the balanced version's low precision (0.180 in the single-split test) means most of its "reversal" predictions are false alarms, and adopting it would require a policy judgment about the relative cost of missed reversals versus false alarms that this paper does not make. It is worth noting that the balancing approach used here, `class_weight="balanced"`, reweights the loss function without reducing the training sample size. An alternative approach, undersampling the majority class to equalize the two classes directly, was not used, but would have shrunk the effective training set substantially — on the order of 400-500 observations, given roughly 196 reversals in the ~1,234-observation training set used in Section 4.2 — likely too few observations for either model to train reliably. This is a further reason the class-weighting approach, rather than undersampling, was the more reasonable choice to test in the first place.

5.3 Statistical and methodological limitations

The Random Forest results reported here come from a single model fit per specification (`random_state=0`). Because Random Forest is sensitive to its bootstrap resampling and random feature subsampling, the numbers reported in Section 4 may vary somewhat under a different seed; no such stability check is reported here. This does not apply to the logit specification, whose optimization is deterministic given a fixed solver and dataset. A more complete treatment would average Random Forest results over multiple seeds.

Standard errors, p-values, and formal statistical inference for the logit coefficients are also absent from this paper; the coefficients reported (Section 4.4) should be read as point estimates rather than statistically tested effects. Adding a formal inference step — for example, fitting the logit specification in `statsmodels` to obtain standard errors and significance tests — is a natural addition for future work.

Finally, the choice not to include country or quarter fixed effects (Section 3.3) was not tested directly against a fixed-effects alternative within this paper. Earlier testing in this project's development suggested country fixed effects contributed negligible predictive value once training data was balanced, but this finding predates the corrected `reserves_pct_gdp` variable and Random Forest design used here, and has not

been re-verified under the final specifications reported.

5.4 What would be done differently

- Assign an explicit cost to a missed reversal versus a false alarm to decide between the balanced and unbalanced specification (Section 5.2), rather than leaving it as an open trade-off.
- Report Random Forest results averaged across multiple random seeds rather than a single fit, given its sensitivity to random resampling.
- Test the model at a different data frequency (e.g., monthly or even annual), which may ease the coverage gaps and warm-up constraint tied to the quarterly design (Section 2.3) — though results would likely differ by frequency, and the right choice depends on the intended use of the early-warning system.
- Test a country fixed-effects specification directly against the non-fixed-effects models used throughout this paper (Section 3.3), rather than leaving that trade-off unexplored.
- Test a multinomial logit (tranquil/pre-crisis/crisis/post-crisis states; Bussière & Fratzscher, 2006) against the binary specification used here, to check whether the binary framing loses information about the build-up phase of a reversal.
- Revisit LASSO variable selection against a broader candidate set of push/pull indicators, and test additional models (XGBoost; the isolation-forest anomaly-detection approach of Griebsch & Röthig, 2026) not covered in this paper.
- Compare results across countries directly, rather than only pooling all 23 countries together, to check whether performance and coefficient patterns are consistent across the panel or driven by a subset of countries.

6. Conclusion

This paper built and evaluated a quarterly early-warning model for portfolio investment reversals across 23 emerging market economies, comparing logistic regression against a Random Forest under a genuinely out-of-sample, forward-chaining design. Both models capture modest discriminative signal: walk-forward AUROC sits only slightly above 0.5 for both, and neither outperforms the naive baseline on a harder single-split test window. Random Forest, which uses class-weight balancing, achieves a better balance between recall and false alarms than the primary (unbalanced) logit, which favors accuracy at the cost of catching few true reversals. A balanced logit was also tested and achieved even better recall and F1 than either primary specification, but at a substantial accuracy cost; this paper reports the unbalanced logit as primary and treats the balanced alternative as a documented trade-off. Both models agree that global push factors — particularly US monetary policy and oil prices — are central to reversal risk, consistent with the "global financial cycle" framing motivating this paper's variable selection; both also identify `reserves_pct_gdp` as highly influential, though its coefficient sign runs counter to theoretical expectations and is not resolved here.

The modest performance reported here should also be read against what the model was, and was not, able to include: variables such as external debt ratios or measures of political stability, which the broader early-warning literature treats as potentially

important, could not be consistently sourced across all 23 countries and are absent from this paper's predictor set (Section 2.2). It is possible that some of the weak discriminative power documented here reflects this gap in available variables, rather than a fundamental limitation of the logit or Random Forest approach itself.

Taken as a whole, these results suggest that a simple logit or Random Forest specification, built on standard push and pull factors, offers limited but non-trivial value as an early-warning tool for this outcome, rather than either a strong success or an outright failure. The gap between what a single evaluation window suggests and what the fuller forward-chaining evaluation shows is itself a useful finding: it illustrates why evaluation design matters as much as model choice when working with a rare, time-ordered outcome. The limitations discussed in Section 5 — incomplete data coverage, an unresolved coefficient puzzle, and a narrow model space — mean these results should be read as a modest first step rather than a finished early-warning system, and the changes outlined there would be the natural priorities for extending this work.

References

Bussière, M., & Fratzscher, M. (2006). "Towards a New Early Warning System of Financial Crises." *Journal of International Money and Finance*, 25(6), 953-973.

Breiman, L. (2001). "Random Forests." *Machine Learning*, 45(1), 5-32.

Calvo, G. A., Izquierdo, A., & Mejía, L. F. (2004). "On the Empirics of Sudden Stops: The Relevance of Balance-Sheet Effects." *NBER Working Paper No. 10520*.

Farnham, S., & Tiftik, E. (2018). *Introducing EM Portfolio Flows Tracker 4.0*. IIF Methodology Note.

Frankel, J., & Saravelos, G. (2012). "Can Leading Indicators Assess Country Vulnerability? Evidence from the 2008-09 Global Financial Crisis." *Journal of International Economics*, 87(2), 216-231.

Griebsch, S., & Röthig, A. (2026). "Bivariate Sudden Stop Analysis of Equity and Bond Fund Flows to Emerging Markets Using Isolation Forest." Deutsche Bundesbank Discussion Paper.

Jarmulska, B. (2022). "Random Forest Versus Logit Models: Which Offers Better Early Warning of Fiscal Stress?" *Journal of Forecasting*, 41(3).

Koepke, R., & Paetzold, S. (2020). *Capital Flow Data – A Guide for Empirical Analysis and Real-Time Tracking*. IMF Working Paper No. 20/171.

Rey, H. (2013). "Dilemma not Trilemma: The Global Financial Cycle and Monetary Policy Independence." *Proceedings – Economic Policy Symposium – Jackson Hole*, Federal Reserve Bank of Kansas City, pp. 285-333.